

Linear Regression Model

Alejandro Melo Ponce



Stony Brook University

Case Study

Class Size and Students' Performance

- What is the effect of reducing class size (measured by *STR*) on students' performance (measured by the average test score on a standardized exam)?
- To answer this question we need a quantitative statement about the relationship between the change in class size and the change in students' performance.
- We can write this relationship as

$$\beta_{ClassSize} = \frac{\text{change in TestScore}}{\text{change in ClassSize}} = \frac{\Delta \text{TestScore}}{\Delta \text{ClassSize}}. \quad (1)$$

Case Study

Class Size and Students' Performance

- If we knew the value of $\beta_{ClassSize}$ we would be able to answer our empirical question since

$$\Delta TestScore = \beta_{ClassSize} \times \Delta ClassSize. \quad (2)$$

- $\beta_{ClassSize}$ is the definition of the slope of a straight line relating test scores and class size

$$TestScore = \beta_0 + \beta_{ClassSize} \times ClassSize. \quad (3)$$

Knowing the values of β_0 and $\beta_{ClassSize}$ would allow us to determine not only the change in the test score due to a change in class size, but also the predicted test score itself for a given class size.

Case Study

Class Size and Students' Performance

- However, there are other variables affecting test scores, and thus the relationship above should be interpreted as a relationship that holds on average.
- A relationship that holds for each observation (district in this case) must incorporate these other factors affecting test scores. For now, we will simply aggregate all these factors under “other factors”

$$TestScore = \beta_0 + \beta_{ClassSize} \times ClassSize + \text{other factors.} \quad (4)$$

- Unfortunately, we don't know the value of the population parameters β_0 and $\beta_{ClassSize}$. As we did for the population mean in chapter 3, we will have to estimate their values using sample data.

The Population Regression Model

Some Notation

- Suppose we have a sample of n school districts. Let Y_i be the average test score in the i -th district, let X_i be the average class size in the i -th district, and let u_i denote all “other factors” in the i -th district. Then we can write the population relationship as

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad \forall i = 1, \dots, n. \quad (5)$$

- The slope of the population regression line is the expected effect on Y of a unit change in X .
- Ultimately our aim is to estimate the causal effect on Y of a unit change in X but for now, just think of the problem of fitting a straight line to data on two variables, Y and X .

The Population Regression Model

Linear Regression Model With One Regressor

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad \forall i = 1, \dots, n.$$

- We have n observations (X_i, Y_i) .
- X is the *independent* variable and Y is the *dependent* variable.
- β_0 is the intercept and β_1 the slope.
- u_i is the regression error, and it consists of omitted factors. In general, these omitted factors are other factors that influence Y , other than the variable X . The regression error also includes error in the measurement of Y .

The Population Regression Model

Linear Regression Model With One Regressor

- The problem of statistical inference for linear regression is, at a general level, the same as for estimation of the mean.
- Statistical, or econometric, inference about the slope entails:
 - Estimation
 - Hypothesis testing
 - Confidence intervals

The Population Regression Model

Estimation

- How should we draw a line through the data to estimate the population slope? *Answer:* ordinary least squares (*OLS*).
- What are advantages and disadvantages of *OLS*?

Hypothesis Testing

- How to test if the slope is not zero? (i.e., the population coefficient is statistically significant).

Confidence Intervals

- How to construct a confidence interval for the slope?

The Population Regression Model

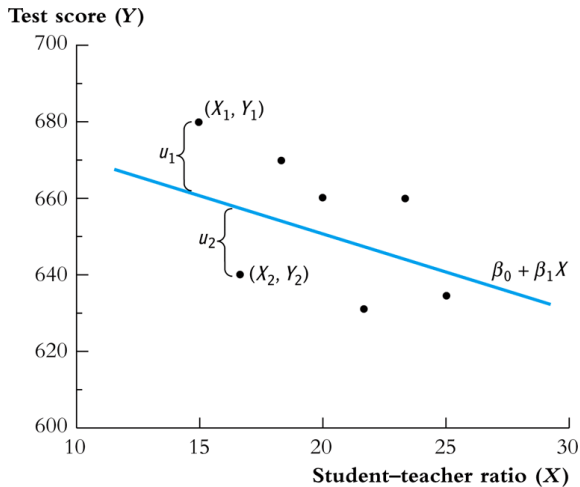


Figure : Linear Regression

Ordinary Least Square Estimator

The OLS Estimator

- Let b_0 and b_1 be some estimators of β_0 and β_1 . Then, the regression line based on these estimators is $b_0 + b_1X$, so the value of Y_i predicted using this line is $b_0 + b_1X_i$.
- Thus, the mistake made in predicting the i -th observation is $Y_i - (b_0 + b_1X_i) = Y_i - b_0 - b_1X_i$, and the sum of the squared mistakes is $\sum_{i=1}^n (Y_i - b_0 - b_1X_i)^2$.
- The *OLS* estimators of β_0 and β_1 are the expressions for b_0 and b_1 that minimize the sum of squared errors above.
- We will denote these estimators by $\hat{\beta}_0$ and $\hat{\beta}_1$.

Ordinary Least Square Estimator

The OLS Estimator

- The *OLS* estimator minimizes the average squared difference between the actual values of Y_i and the prediction (“predicted value”) based on the estimated line.
- This minimization problem can be solved using calculus (See: Appendix 4.2 in the textbook).
- The result are the *OLS* estimators of β_0 and β_1 .

Ordinary Least Square Estimator

The OLS Estimators, Predicted Values, and Residuals

The *OLS* estimators of the slope β_1 and the intercept β_0 are

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{s_{XY}}{s_X^2}, \quad (6)$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}. \quad (7)$$

The *OLS* predicted values $\hat{Y}_i$ and residuals $\hat{u}_i$ are

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i \quad i = 1, \dots, n, \quad (8)$$

$$\hat{u}_i = Y_i - \hat{Y}_i \quad i = 1, \dots, n. \quad (9)$$

The estimated $\hat{\beta}_0$, $\hat{\beta}_1$, $\hat{u}_i$ are computed from a sample of n observations (X_i, Y_i) , for $i = 1, \dots, n$. These are estimates of the unknown true population intercept β_0 , slope β_1 , and error term u_i .

Some Algebraic facts about the OLS estimator

- $\frac{1}{n} \sum_{i=1}^n \hat{u}_i = 0$.
- $\frac{1}{n} \sum_{i=1}^n \hat{Y}_i = \bar{Y}$.
- $\sum_{i=1}^n \hat{u}_i X_i = 0$, $s_{\hat{u}X} = 0$.
- $\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 + \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$
- where $TSS = \sum_{i=1}^n (Y_i - \bar{Y})^2$, $SSR = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$ and $ESS = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$

Ordinary Least Square Estimator

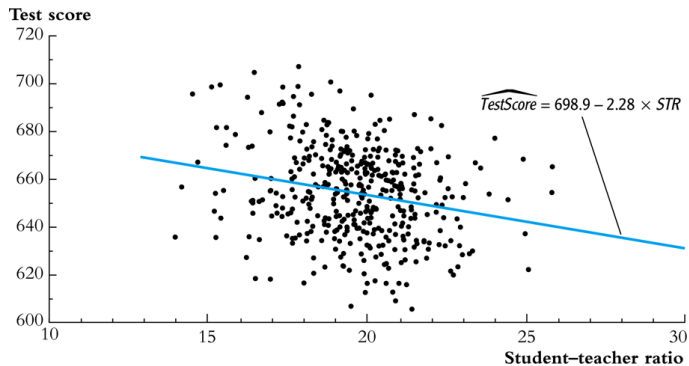


Figure : Linear Regression Estimate

Ordinary Least Square Estimator

Interpretation of the Estimated Slope and Intercept

- The predicted test score is $698.9 - 2.28STR$. Thus, districts with one more student per teacher on average have test scores that are 2.28 points lower. That is

$$\frac{\text{change in TestScore}}{\text{change in ClassSize}} = -2.28.$$

- The intercept (taken literally) means that, according to this estimated line, districts with zero students per teacher would have a (predicted) test score of 698.9 which makes no sense – it extrapolates the line outside the range of the data.

Predicted Values & Residuals

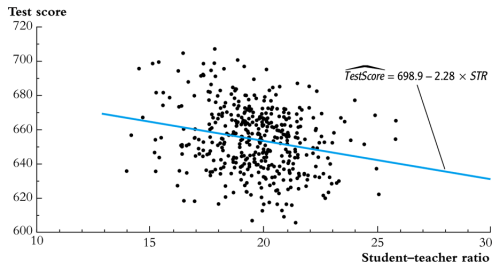


Figure : Linear Regression Estimate

One of the districts in the data set is Antelope, CA, for which $STR = 19.33$ and $TestScore = 657.8$, then

$$\hat{Y}_{Antelope} = 698.9 - 2.28 \times 19.33 = 654.8 \quad (10)$$

$$\hat{u}_{Antelope} = 657.8 - 654.8 = 3. \quad (11)$$

OLS Regression: *Standard output from statistical software*

Number of obs = 420
F(1, 418) = 19.26
Prob > F = 0.0000
R-squared = 0.0512
Root MSE = 18.581

		Robust				
testscr	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	

str	-2.279808	.5194892	-4.39	0.000	-3.300945	-1.258671
cons	698.933	10.36436	67.44	0.000	678.5602	719.3057

Measures of Fit

- Having estimated a linear regression, you might wonder how well that regression line describes the data.
- Does the regressor account for much or for little of the variation in the dependent variable?
- Are the observations tightly clustered around the regression line, or are they spread out?
- Two regression statistics provide complementary measures of how well the regression line “fits” or explains the data: the regression R^2 and the standard error of the regression SER .

Measures of Fit

The Regression R^2

Measures the fraction of the variance of Y that is explained by X ; it is unitless and ranges between zero (no fit) and one (perfect fit).

The Standard Error of the Regression SER

Measures the magnitude of a typical regression residual in the units of Y .

A detour...

- Recall that earlier, we established

$$\begin{array}{rcl} \sum_{i=1}^n (Y_i - \bar{Y})^2 & = & \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 + \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 \\ TSS & = & SSR + ESS \end{array}$$

- TSS is the total variance of Y .
- SSR is the variance of Y **not** explained by X .
- ESS is the variance of Y explained by X .
- Actually, this expression is the sample version of the **law of total variance**.
- $Var(Y) = E[Var(Y|X)] + Var(E[Y|X])$

Measures of Fit

The Regression R^2

- From the definition of predicted value we can write

$$Y_i = \hat{Y}_i + \hat{u}_i. \quad (12)$$

- In this notation, the R^2 is the ratio of the sample variance of $\hat{Y}_i$ to the sample variance of Y_i .
- Mathematically, the R^2 can be written as the ratio of the explained sum of squares (ESS) to the total sum of squares (TSS).

Measures of Fit

The Regression R^2

- From the definition of predicted value we can write

$$ESS = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 \quad TSS = \sum_{i=1}^n (Y_i - \bar{Y})^2 \quad (13)$$

- Therefore, the R^2 is

$$R^2 = \frac{ESS}{TSS}. \quad (14)$$

Measures of Fit

The Regression R^2

- Alternatively, the R^2 can be written in terms of the fraction of the variance of Y_i not explained by X_i .
- The sum of squared residuals (SSR) is

$$SSR = \sum_{i=1}^n \hat{u}_i^2. \quad (15)$$

- It can be shown that $TSS = ESS + SSR$, and then

$$R^2 = 1 - \frac{SSR}{TSS}. \quad (16)$$

Measures of Fit

Remarks about R^2

- $R^2 = 0$ means $ESS = 0$.
- $R^2 = 1$ means $ESS = TSS$.
- $0 \leq R^2 \leq 1$.
- For regression with a single X , R^2 is equal to the correlation coefficient between X and Y .

Measures of Fit

The Standard Error of the Regression *SER*

- The *SER* measures the spread of the distribution of u . The *SER* is (almost) the sample standard deviation of the *OLS* residuals:

$$\begin{aligned} SER &= s_{\hat{u}} \\ &= \sqrt{\frac{1}{n-2} \sum_{i=1}^n \left(\hat{u}_i - \bar{\hat{u}} \right)^2}, \\ &= \sqrt{\frac{1}{n-2} \sum_{i=1}^n \hat{u}_i^2}. \end{aligned} \tag{17}$$

- The second equality comes from the fact that $\bar{\hat{u}} = \frac{1}{n} \sum_{i=1}^n \hat{u}_i = 0$.

Measures of Fit

Remark *SER*

- The *SER* has the units of u , which are the units of Y and measures the average “size” of the *OLS* residual (the average “mistake” made by the *OLS* regression line).
- The *root mean squared error* (*RMSE*) is closely related to the *SER*

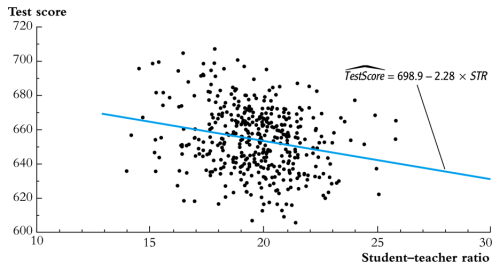
$$RMSE = \sqrt{\frac{1}{n} \widehat{u}_i^2}. \quad (18)$$

Technical Note: why we divide by $n - 2$ instead of $n - 1$

$$SER = s_{\hat{u}} = \sqrt{\frac{1}{n-2} \sum_{i=1}^n (\hat{u}_i - \bar{\hat{u}})^2} = \sqrt{\frac{1}{n-2} \sum_{i=1}^n \hat{u}_i^2}.$$

- Division by $n - 2$ is a “degrees of freedom” correction – just like division by $n - 1$ in s_Y^2 , except that for the SER , two parameters have been estimated β_0 and β_1 (by $\hat{\beta}_0$ and $\hat{\beta}_1$) whereas in s_Y^2 only one parameter has been estimated (μ_Y by $\bar{Y}$).
- When n is large, it does not matter whether n , $n - 1$, or $n - 2$ are used – although the conventional formula uses $n - 2$ when there is a single regressor. For details, see Section 17.4 in the textbook.

Example of the R^2 and SER



$$\widehat{TestScore} = 698.9 - 2.28 \times STR, \quad R^2 = 0.05, \quad SER = 18.6$$

STR explains only a small fraction of the variation in test scores. Does this make sense? Does this mean the STR is unimportant in a policy sense?

Least Squares Assumptions

Least Square Assumptions

- What, in a precise sense, are the properties of the sampling distribution of the OLS estimator? When will $\hat{\beta}_1$ be unbiased? What is its variance?
- To answer these questions, we need to make some assumptions about how Y and X are related to each other, and about how they are collected (the sampling scheme).
- These assumptions –there are three– are known as the Least Squares Assumptions.

Least Squares Assumptions

Least Square Assumptions

Recall the population regression model

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad \forall i = 1, \dots, n.$$

- 1 **The conditional distribution of u_i given X_i has mean zero.** That is, $E[u_i | X_i = x] = 0$.
- 2 **(X_i, Y_i) , $i = 1, \dots, n$, are i.i.d.** This assumption is true if (X, Y) are collected by simple random sampling.
- 3 **Large outliers in X and/or Y are rare.** Technically, X and Y have finite fourth moments. Outliers can result in meaningless values of $\hat{\beta}_1$.

Least Squares Assumption

Assumption 1: $E[u_i|X_i] = 0$. For any value of X the mean of u is zero.

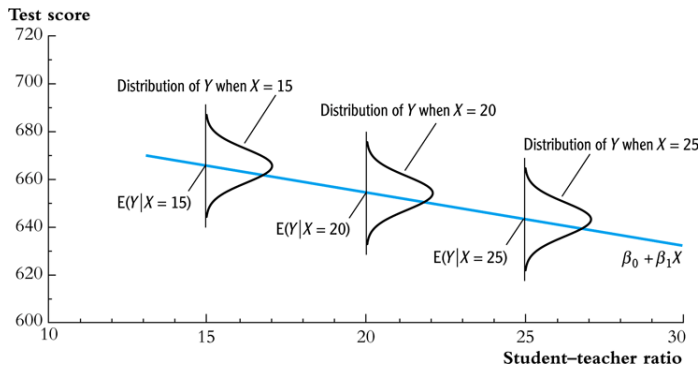


Figure : OLS Assumption 1

Least Squares Assumption

Assumption 1: $E[u_i|X_i] = 0$. For any value of X_i the mean of u_i is zero.

- This assumption is a formal statement about the “other factors” contained in u_i . It means that these other factors are unrelated to X_i .
- A benchmark for thinking about this assumption is to consider an ideal randomized controlled experiment: X is randomly assigned to people (students randomly assigned to different size classes; patients randomly assigned to medical treatments). Randomization is done by computer – using no information about the individual.
- Assumption 1 is equivalent to assuming that the population regression line is the conditional mean of Y_i given X_i .

Least Squares Assumption

Assumption 1: $E[u_i|X_i] = 0$. For any value of X_i the mean of u_i is zero.

- Because X is assigned randomly, all other individual characteristics –the things that make up u – are distributed independently of X , so u and X are independent.
- Thus, in an ideal randomized controlled experiment, $E[u_i|X_i] = 0$ (that is, Assumption 1 holds).
- In actual experiments, or with observational data, we will need to think hard about whether $E[u_i|X_i] = 0$ holds.

Least Squares Assumption

Assumption 2: (X_i, Y_i) are i.i.d.

- This assumption holds automatically if the entity (individual, district, etc.) is sampled by simple random sampling.
- The entities are selected from the same populations, so (X_i, Y_i) are identically distributed for all $i = 1, \dots, n$.
- The entities are selected at random, so the values of (X, Y) for different entities are independently distributed.
- We can expect to encounter non-i.i.d. data when information is recorded over time for the same entity (panel data and time series data).

Least Squares Assumption

Assumption 3: Large Outliers are Rare.

- Formally, this is obtained by assuming that X_i and Y_i have nonzero finite fourth moments, i.e., $0 < E[X_i^4] < \infty$ and $0 < E[Y_i^4] < \infty$, or in other words, that the distributions of X_i and Y_i have finite kurtosis.
- Look at your data! If you have a large outlier, is it a typo? Does it belong in your data set? Why is it an outlier?
- In practice, outliers are often data glitches (coding or recording problems). Sometimes they are observations that really shouldn't be in your data set.
- Plot your data!

Least Squares Assumption

Assumption 3: Large Outliers are Rare

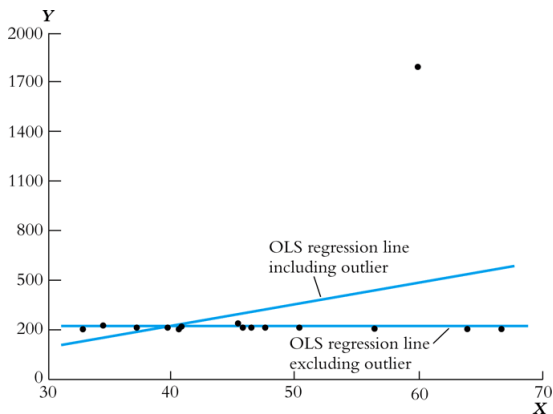


Figure : OLS Assumption 3

Least Squares Assumption

Use of the *OLS* Assumptions.

- If the assumptions hold and the sample size is large, the *OLS* estimators have sampling distributions that are normal, and therefore we can conduct hypothesis test about the unknown population parameters based on the sample estimates.
- The assumptions also restrict the use of the *OLS* estimators.

Sampling Distribution of the *OLS* Estimator

The *OLS* estimator $\hat{\beta}_1$ is computed from a sample of data. A different sample yields a different value of $\hat{\beta}_1$. This is the source of the “sampling uncertainty” of $\hat{\beta}_1$.

We want to:

- quantify the sampling uncertainty associated with $\hat{\beta}_1$,
- use to test hypotheses such as $\beta_1 = 0$.
- construct a confidence interval for β_1 .

All these require figuring out the sampling distribution of the *OLS* estimator.

Sampling Distribution of the *OLS* Estimator

Large Sample Distribution of $\hat{\beta}_1$ and $\hat{\beta}_0$.

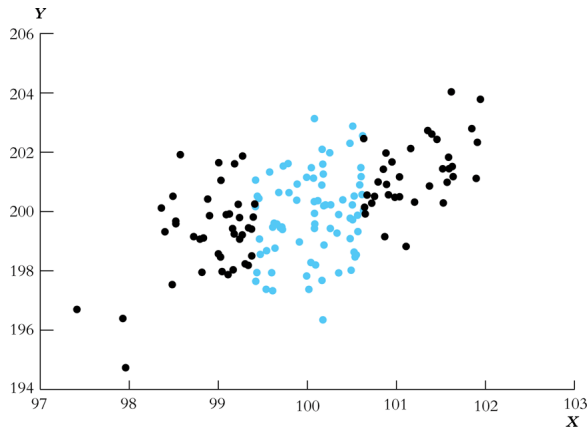
If the three *OLS* assumptions hold and the sample size n is large, $\hat{\beta}_1$ and $\hat{\beta}_0$ are jointly normally distributed. The distribution of $\hat{\beta}_1$ is $N(\beta_1, \sigma_{\hat{\beta}_1}^2)$ where

$$\sigma_{\hat{\beta}_1}^2 = \frac{1}{n} \frac{\text{Var}[(X_i - \mu_X)u_i]}{(\text{Var}[X_i])^2}. \quad (19)$$

The distribution of $\hat{\beta}_0$ is $N(\beta_0, \sigma_{\hat{\beta}_0}^2)$ where

$$\sigma_{\hat{\beta}_0}^2 = \frac{1}{n} \frac{\text{Var}[H_i u_i]}{(E[H_i^2])^2}, \quad \text{where } H_i = 1 - \frac{\mu_X}{E[X_i^2]} X_i. \quad (20)$$

Sampling Distribution of the *OLS* Estimator



The blue and black samples have the same size. Which one do you think would give you a better prediction for the population line?